

LOLM

Latent Order Language Model

A Hybrid Transformer-SSM Architecture with Learned Gating,
Discrete Regime Detection, and Persistent Memory

Bryan Leonard & Brandyn Leonard
Qira LLC | Patent App #64/002,166

THE PROBLEM

Standard Transformer decoders treat language as a flat sequence of tokens.
But language has hidden structure that pure attention cannot efficiently capture.

1 Fixed Context Window

Attention can't track dynamics
beyond its window length

2 No Latent State

No persistent representation
of discourse-level variables

3 Scale Inefficiency

Bigger models don't learn
qualitatively different features

THE LOLM ARCHITECTURE

Five parallel streams converge through learned per-dimension gating



Surface Decoder

Pre-norm Transformer + RoPE



Latent SSM Core

Mamba-style selective scan



Regime Layer

Gumbel-Softmax phase detection



Persistent Memory

3-bank episodic/semantic/self



Manifestation Gate

Per-dim surface/latent blend

KEY INNOVATION

The gate learns ~29% latent signal is indispensable. Removing it causes a 14,000,000x PPL explosion.

The minority path becomes MORE critical at scale -- dependency inversion.

PROVEN RESULTS

15%

Better PPL at 1.57B
H200 vs matched baseline

52%

Better vs Pythia-410M
304M, 26% fewer params

14Mx

Gate Ablation Explosion
Removing latent path

7.6%

Better PPL on TPU
Cross-hardware validation

Scale	LOLM	Baseline	Delta
1.57B (H200)	PPL 33.2	PPL 39.1	+15.1%
304M vs Pythia	PPL 68.4	PPL 142.9	+52.2%
300M (TPU matched)	PPL 380	PPL 411	+7.6%
300M (param-matched)	PPL 432 @15K	PPL 443 @15K	+2.4%

DOWNSTREAM EVALUATION

Benchmark	FULL LOLM	Baseline	Pythia
WikiText-103 PPL	2,026	2,166	~30
HellaSwag	26.0%	22.0%	30.5%
LAMBADA	1.0%	1.0%	~5%

Cross-Hardware Validation

H200 (CUDA): 15% better at 1.57B

TPU v4 (XLA): 7.6% better at 300M

3 datasets: FineWeb-Edu, C4, Pile

Scales with Model Size

20.5M -> 149M -> 304M -> 1.57B

Dependency inversion intensifies
at scale -- latent path MORE critical

WHAT WE NEED

1 GPU Access

8x H100/H200 for 90 days
to train 7B FULL LOLM

2 Technical Mentorship

NVIDIA AI experts to
optimize kernels + FLOPs

3 Publication Support

Co-publication at
NeurIPS/ICML 2026-2027

WHAT NVIDIA GETS

Novel architecture making NVIDIA GPUs more efficient. LOLM = equivalent quality, fewer parameters, less compute, less cost per token. Published NVIDIA advantage.

TEAM & IP

Bryan Leonard

Co-Founder & Lead Architect

LOLM architecture, training framework, full-stack ML

Brandyn Leonard

Co-Founder & Research Lead

Theoretical foundations, experimental validation

Protected IP

Patent: App #64/002,166 (Mar 2026)

License: LOLM Community License v1.0

Code: github.com/TheArtOfSound/LOLM

Website: qira.dev

LOLM proves that language modeling benefits from explicitly separating surface and latent representations -- and the benefit grows with scale.

bryanleonard237@gmail.com
github.com/TheArtOfSound/LOLM
qira.dev

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